

Case Study Problem - NumPy

1. Create a function that takes dimensions as tuples e.g.(3, 3) and a numeric value and returns a Numpy array of the given dimension filled with the given value.

PROGRAM

```
import numpy as np

def tup():
    y=[]
    for i in range(2):
        y.append(int(input()))
    a=tuple(y)
    b=np.full(a,6)
    return b
print(tup())
```

INPUT

3
2

OUTPUT

```
[[6 6]
 [6 6]
 [6 6]]
```

2. Create a method that takes n Numpy arrays of same dimensions sums them and return the answer.

PROGRAM

```
import numpy as np

def add1(c,d):
    f=np.add(c,d)
    return f

c=np.array([1,2,3])
d=np.array([4,5,6])

print(add1(c,d))
```

OUTPUT

```
[5 7 9]
```

3. Given a 2 D Array of N X M Dimension, Write a function that accepts this array as well as two numbers N and M. The method should return the top-left N X M sub matrix.

PROGRAM

```
import numpy as np

matrix=np.array([1,2,3,4,5,6,7,8,9])
x=matrix.reshape(3,3)

def top_left(matrix,a,b):
    print(matrix[:a,:b])

top_left(x,2,2)
```

OUTPUT

```
[[1 2]
 [4 5]]
```

4. Given a 2 D Array of N X M Dimension, Write a function that accepts this array as well as two numbers N and M The method should return the bottom-right N X M sub matrix.

PROGRAM

```
import numpy as np

matrix=np.array([1,2,3,4,5,6,7,8,9])
x=matrix.reshape(3,3)

def bottom_right(matrix,a,b):
    print(matrix[a:,b:])

bottom_right(x,1,1)
```

OUTPUT

```
[[5 6]
 [8 9]]
```

5. Given a 1 D Numpy Array. Write a function that accepts this array as parameters. The method should return a dictionary with 'mean' and 'std_dev' as key and array's mean and array's standard deviation as values.

PROGRAM

```
import numpy as np

def compute(k):
    m=np.mean(k)
    n=np.std(k)

    dict={"mean":m,"std":n}
    return dict

k=np.array([3,1,5,8,4])

print(compute(k))
```

OUTPUT

```
{'mean': 4.2, 'std': 2.3151673805580453}
```
